

Indian mythology mentions *Vimanas*, a spacecraft belonging to Ravana and Salva

Propulsion engines
That breathe air and will save the world

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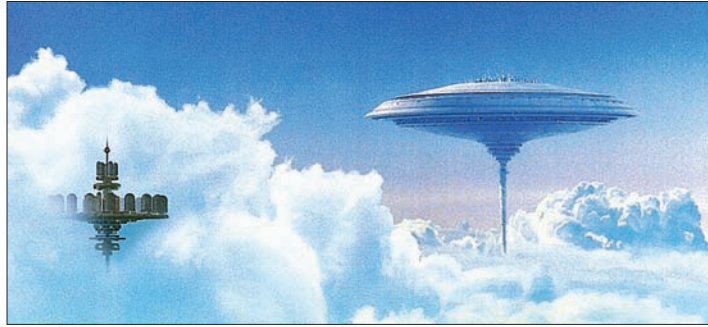
for a cable stretching from the Earth to a counterweight at a geosynchronous orbit.

Beyond 2500

The Kardashev scale is a measure of the energy consumption of a civilisation. A type I civilisation can harness the entire energy from a planet. Similarly, a type II civilisation can harness the energy output of a star, while a type III civilisation, the energy output of an entire galaxy. At present, humans are at 0.7, as in we have not yet been able to harness the entire energy output of the planet. Conversely, the Kardashev scale can also show the technical prowess of a civilisation. With a lot of energy being consumed, humans can manage drastic changes to the environment.

Jupiter, will be an important planet that mankind will be looking at. A 2003 study by NASA about the future of colonisation, suggested Callisto as a candidate. Callisto is an ideal location to serve as a fuel stop for exploration into the deeper parts of the Solar System, beyond Jupiter. Jupiter itself is rich in carbon compounds and also helium. Cities in the sky, using large gas balloons are also being considered. This approach would also work for permanent bases on Venus. Floating cities are the most likely approaches for planets with a lot of gases in the atmosphere, including Saturn and Venus.

Terraforming is one of the most likely ideas of a post-type I civilisation. This involves changing the entire surface of a planet to make it habitable. Terraforming the Moon or Mars would remove the necessity of an enclosed base, and allow for a vast, Earth-like environment on another planet. This operation is most likely to be exercised on Mars, as the planet has a gravitational field



A proposed model for a sky city

that could retain oxygen, nitrogen and other vital gasses. Venus is another proposed candidate, with cities being established only in the outer atmosphere to begin with, and the terraforming operation eventually clearing up the poisonous skies for human habitation. These processes would take at least five decades to execute, so these activities will be long term projects.

At this juncture, humans will also start seriously looking at interstellar travel. There are a number of approaches at this point of time, the most likely to succeed will be a space ark that takes a breeding population, and several generations to reach the stars. In all probability, a fleet of such arks will be used instead of just one large ship. Other possibilities include cryogenic freezing, where humans are re-animated after a long journey in space.

While the timescales for these events may be off, the eventuality of such measures by mankind are inevitable. The reason for this is simply an instinct that has kept us from going extinct — self-preservation. There are a number of calamities that could befall this planet. The climate might change drastically, or a number of natural disasters, from a major asteroid impact to massive solar flares, could endanger life on Earth. At least a few of us would survive any eventuality, including a full-scale nuclear war, and life will go on. However, there is a clock ticking for the planet and the Sun. The Sun has a life cycle too, albeit a very long one. Eventually, the Sun will expand as it runs out of fuel, and when this event occurs, the Earth will be obliterated. This is why we reach for the stars. ■

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